Landslide Susceptibility Hot Spot Analysis for West Virginia

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Introduction

Landslides represent a large hazard with large socio-economic and environmental impacts, especially in regions with steep topography and vulnerable communities (Alcántara-Ayala, 2025). The rapid movement of soil, rock, and debris down slope is often triggered by intense rainfall, seismic activity, volcanic eruptions, and human-induced alterations to the landscape (Alcántara-Ayala, 2025). Landslides can cause extensive damage and can result in loss of life, disproportionately affecting vulnerable communities and placing significant strain on local and regional governments (Alcántara-Ayala, 2025; Maxwell, 2021). Climate change just makes things worse by altering precipitation patterns and increasing soil moisture variability in areas already prone to instability (Alcántara-Ayala, 2025)

Landslides are not randomly distributed across a space, but tend to show spatial and temporal clustering (Tonini et al., 2014). This clustering seems to be influenced by underlying environmental characteristics like slope, aspect, curvature, soil properties, and land cover, as well as hydrologic factors like surface runoff and subsurface flow (Maxwell, 2020). Because of these characteristics, an understanding of the spatial structure of landslide susceptibility is critical to assess hazard and target funding for areas most prone to landslides.

Spatial autocorrelation provides a statistical measure of how much something is correlated with itself in space and determines spatial patterns like clustering, dispersion, or if it's randomly spaced (Zhang & Zhang, 2015). In general, measures of spatial autocorrelation can be classified as global or local measures (Yeh & Liaw, 2021). Global measures assume that nothing moves and provide a single value that summarizes the entire area (Yeh & Liaw, 2021). Local measures, on the other hand, help identify local patterns of clustering and dispersion, making them especially important in areas where the local trends might differ from the global trend (Yeh & Liaw, 2021). When it comes to global measures, Moran's I is one of the most widely used statistics to test for spatial autocorrelation and is frequently used in studies of land cover and environmental processes (Yeh & Liaw, 2021). This tests whether further analysis is even needed in this study. Getis-Ord General G statistic is used to assess whether high or low landslide susceptibility values were globally clustered, offering additional insight into the data (Zhang & Zhang, 2015).

When studying landslides, hydrologic conditions are important to consider (Yeh & Liaw, 2021). Because of this, watersheds represent one of the best spatial units for this analysis as they help characterize the terrain, drainage, and surface processes (Yeh & Liaw, 2021). Summarizing data into watershed units helps scale the data for regional planning and hazard management.

While global statistics provide important summary information, they fail to capture the scale at which the spatial clustering occurs. Multi-distance spatial statistics, done through Ripley's K Function, is the solution to this problem by testing for clustering for a range of distances (Tonini et al., 2014). Ripley's K helps identify at which scale landslide susceptibility is most strongly clustered, and be able to target the analysis to that scale (Tonini et al., 2014). Combining with a local statistic of spatial autocorrelation, like Getis-Ord Gi* statistics, allows for both the detection of clustering and the statistically significant hot and cold spots to be identified (Ord & Getis, 1995).

This study tests the spatial relationships of landslide susceptibility in West Virginia using global, local, and multi-distance spatial statistics. Specifically, Global Moran's I and Getis-Ord General G are used to test for overall spatial autocorrelation and clustering of hot or low susceptibility values, Ripley's K Function is applied to identify the spatial scales at which clustering occurs, and local Hot Spot Analysis through Getis-Ord Gi* is used to map statistically concentrated areas of high and low susceptibility. By combining these complementary methods within a watershed summarization of landslide susceptibility, this study provides a way to test and determine the spatial patterns of landslide susceptibility across West Virginia.

Methodology

A raster of landslide susceptibility was created as a part of an ongoing project to map hazards for the state of West Virginia (Maxwell, 2021; Maxwell, 2020). This raster was made looking exclusively at LiDAR-based digital elevation model (DEM) variables, allowing for a two-meter by two-meter scale raster to be created (Maxwell, 2021; Maxwell, 2020). While this resolution is good for site-level analysis, identifying broader spatial patterns was difficult at this scale, but resampling the raster to 30-meter by 30-meter resolution didn't provide a large improvement. The solution to better examine regional patterns was to convert the raster of landslide susceptibility into a vector.

For this study, landslide susceptibility was summarized using the HUC–10 watershed, seen in Map 1. Although county-level units are commonly used in the United States, they lack the ability to take into account water and its importance when it comes to landslides. When looking at landslides, the two main factors to take into account are topography and water (Yeh & Liaw, 2021; Alcántara-Ayala, 2025). Watersheds allow for characterizations of both components, taking into account where water will flow, but also the cavities where landslides occur following where the water flows.

Understanding where landslides tend to occur through the mean landslide susceptibility by watershed, seen in Map 2, was the first step. The map shows by its colors already a clustering of roughly four areas: the southwestern point of the state, the central eastern side of the state, just south of the northern panhandle, and just south of the eastern panhandle. While these areas might look like clustering, it's important to be able to identify these hotspots statistically, which will be done through various methods.

Starting at the highest level of understanding, clustering was done through three different methods to understand global measures of clustering (Yeh & Liaw, 2021; Zhang & Zhang, 2015). Global tests are about understanding overall patterns of a large region, which often provide a single value as a result (Yeh & Liaw, 2021; Ord & Getis, 1995). Global autocorrelation indexes can be seen in the following tests, which this study will do: Moran's I, Getis-Ord General G, and Ripley's K (Zhang & Zhang, 2015). The first test performed on the data was a Global Moran's I test, which is widely used to estimate general patterns of spatial dependency (Yeh & Liaw, 2021). This test is also named the test of spatial autocorrelation because it is used to understand how a phenomenon of interest is correlated to itself over a large area (Yeh & Liaw, 2021; Zhang & Zhang, 2015). When performing this test, it was important to note that the relationship of which polygons were deemed to exert influence was changed from the default to "contiguity edges corners." This option in the Global Moran's I test allows for polygon features that share a border, node, or overlap to be determined to have influence on their neighbors. The nature of watersheds being neighbors for the flow of water allows for this option of neighbor selection to make the most sense. The next test was the Getis-Ord General G test, which also used the same method to classify neighbors. General G is used to assess whether high or low landslide susceptibility values were globally clustered (Zhang & Zhang, 2015).

The final global test was Ripley's K function, which is widely used to study spatial patterns and determine the scale at which clustering, random spread, or dispersion is observed (Ord & Getis, 1995; Tonini et al., 2014). This test requires data to be based on points, which was done by using the centroids of polygons (Tonini et al., 2014). Using all the watershed polygon centroids would be meaningless; instead, the z-score of the mean landslide susceptibilities was used. Z-score stands for the number of standard deviations a point is from the mean of the dataset as a whole. The z-scores were calculated for all the watersheds, and only the watersheds with a z-score greater than one were the polygons' centroids used for this analysis. Scales set to be tested were from 10 km for 15 times, resulting in a range from 10 km to 150 km to be tested. The range was determined by using a rough size of a watershed up to almost the size of the state as a whole. The boundary area values were set to be simulated, which allows for simulated points to be created outside the study area so that the number of neighbors near the edges is not underestimated (Tonini et al., 2014). A mirroring effect takes effect at the edges, reflecting the points for calculations based on the mirror of the points inside the boundary. This has to be used to allow for estimations of what the landslide susceptibility values are like in neighboring states. The final decision needed to be made was the number of iterations, which was set to 99, the middle ground between the options in ArcGIS Pro.

The final analysis performed on this data goes forward to try to determine the local-level spatial patterns derived from understanding how individuals are classified in relationships to surrounding neighbors (Yeh & Liaw, 2021). These statistics, like Getis-Ord Gi*, are useful in cases where global statistics may fail to alert the researchers to significant pockets of clustering within the data (Ord & Getis, 1995). Patterns of global statistics often fail to see other useful patterns that can only be seen at a local scale (Ord & Getis, 1995). Getis-Ord Gi* is a z-score-based statistic that is used to test for statistically significant clusters of high and low areas in a study area (Ord & Getis, 1995; Yang et al., 2020; Sultana, 2020). There is a history of using Getis-Ord Gi* to study landslides to understand if landslides show clustering, which studies have shown they do (Yang et al., 2020; Sultana, 2020). Helping to run the Getis-Ord Gi* the relationship of which polygons were deemed to exert influence was changed from the default to “contiguity edges corners,” identical to Getis-Ord General G and Moran’s I. The Gi* results in each watershed being given a z-score, which is then reclassified based on these z-scores to confidence levels.

Results

The results of the global patterns showed that both Moran's I and Getis-Ord General G had strong levels for their values, respectively. Global Moran's I showed a very strong positive spatial autocorrelation in the landslide susceptibility values across HUC-10 watersheds ($I = 0.72$, $z = 15.19$, $p < 0.001$), showing that watersheds with similar susceptibility tend to be spatially clustered. Having Moran's I value near a positive one indicates clustering, seen in this test with $I = 0.72$, which is further supported by results seen in the General G (Yeh & Liaw, 2021). The Getis-Ord General G indicated significant global clustering of high landslide susceptibility values ($z = 6.26$, $p < 0.001$), suggesting that areas of high susceptibility are regionally clustered rather than randomly distributed. While General G tests for clustering as a whole, either for low value clustering or high value clustering, the positive z-score indicates a higher degree of high value clustering.

The strong global spatial autocorrelation identified by Moran's I and the clustering of high values indicated by General G are consistent with the results of Ripley's K, seen in Figure 1. The graph shows that statistically significant clustering of high landslide susceptibility for watersheds occurs at distances ranging from approximately 30 km to 140 km. The graph shows these results by looking at where the Observed K is above the upper confidence envelope (Tonini et al., 2014). This pattern suggests that landslide-prone areas in West Virginia are not randomly distributed but instead show regional-scale clustering. The patterns at the smaller scale, below 30 km, and at larger scales, above 140 km, indicate patterns that approach spatial randomness.

Lastly, the Getis-Ord Gi* results can be seen in Map 3. The map is divided evenly into two clustering groups for both hot and cold spots. The hot spots, which are clusters of high landslide susceptibility values surrounded by more of the same, are seen on the east side of the state primarily. The southeastern point of the state has a large cluster of seven watersheds, which have a hot spot with 99% confidence. This cluster then also has five watersheds with a 95% confidence surrounding them, and two with 90% confidence beyond them. With only a single watershed between them, the other cluster of hot spots, with 99% confidence, has a large cluster of nine watersheds located just south of the northern panhandle. This second cluster also has six watersheds with 95% confidence surrounding them, and six with 90% confidence. Apart from both these hot spots, there is just one area that has cold spots with 99% confidence, being the eastern panhandle, with thirteen watersheds. This area then proceeds to only have two with 95% confidence and five with 90% confidence surrounding them. This shows a more pronounced clustering of cold spots, given that there are more watersheds with 99% and fewer surrounding them of 95% and 90%. The second cold spot of note only has two watersheds, with 95% surrounded by six watersheds, with 90% in the southern interior part of the state.

Conclusion and Discussion

While the original raster of landslide susceptibility was great for understanding susceptibility at the two-meter scale, using watersheds for this analysis allows for an understanding of broader patterns in the data. Using the HUC-10 watershed for West Virginia allowed for global, multi-distance, and local spatial statistics to be calculated. Results consistently showed that landslide susceptibility is not randomly distributed, but instead shows strong spatial dependence and regional clustering across the state.

Global Moran's I showed every strong positive spatial autocorrelation, which indicates that watersheds with similar susceptibility values tended to be near one another. This suggests that continuous regional variations in things like slope, soil properties, land cover, vegetation, and precipitation patterns, rather than isolated local factors, play a dominant role in landslide susceptibility (Maxwell, 2021; Maxwell, 2020). These findings are supported by the way the original raster was created through the use of just LiDAR DEM-derived variables, which could be a limitation of using this analysis to truly understand landslide susceptibility clustering risk (Maxwell, 2021; Maxwell, 2020). Getis-Ord General G analysis goes further in showing that high landslide susceptibility values are globally clustered, rather than evenly distributed throughout the state. Together, the Morans' I and General G results suggest that landslide susceptibility in West Virginia is both spatially structured overall and characterized by regional clustering of high-risk areas.

Ripley's K Function brought insight to the scale at which clustering occurs, showing that statistically significant clustering of high-susceptibility watersheds occurs at distances between 30 km and 140 km. This distance range shows that the landslide susceptibility pattern of clustering occurs at a regional scale, rather than being impacted by local factors.

Getis-Ord Gi* furthers the analysis to a local scale that helped identify statistically significant hot and cold spots of landslide susceptibility. The cold spots show areas where development could be deemed less hazardous, in comparison to the lessened risk of landslides for the state as a whole. Hot spots show where watersheds with high landslide susceptibility are surrounded by other watersheds with high landslide susceptibility. These areas should be areas where funding should be focused due to the increased risk of landslides. These risks are likely to intensify if climate change continues to alter precipitation patterns and soil moisture levels that can trigger landslides in areas already prone to instability (Alcántara-Ayala, 2025).

In conclusion, this study shows that landslide susceptibility in West Virginia is strongly spatially autocorrelated, regionally clustered, and scale-dependent. The combined use of Global Moran's I, Getis-Ord General G, Ripley's K Function, and Getis-Ord Gi* provides a comprehensive group of tests to understand both the intensity and spatial organization of landslide susceptibility. Future research could be done to work with landslide inventories to validate this clustering or test various scales to see if clustering is still seen in similar areas.

Sources

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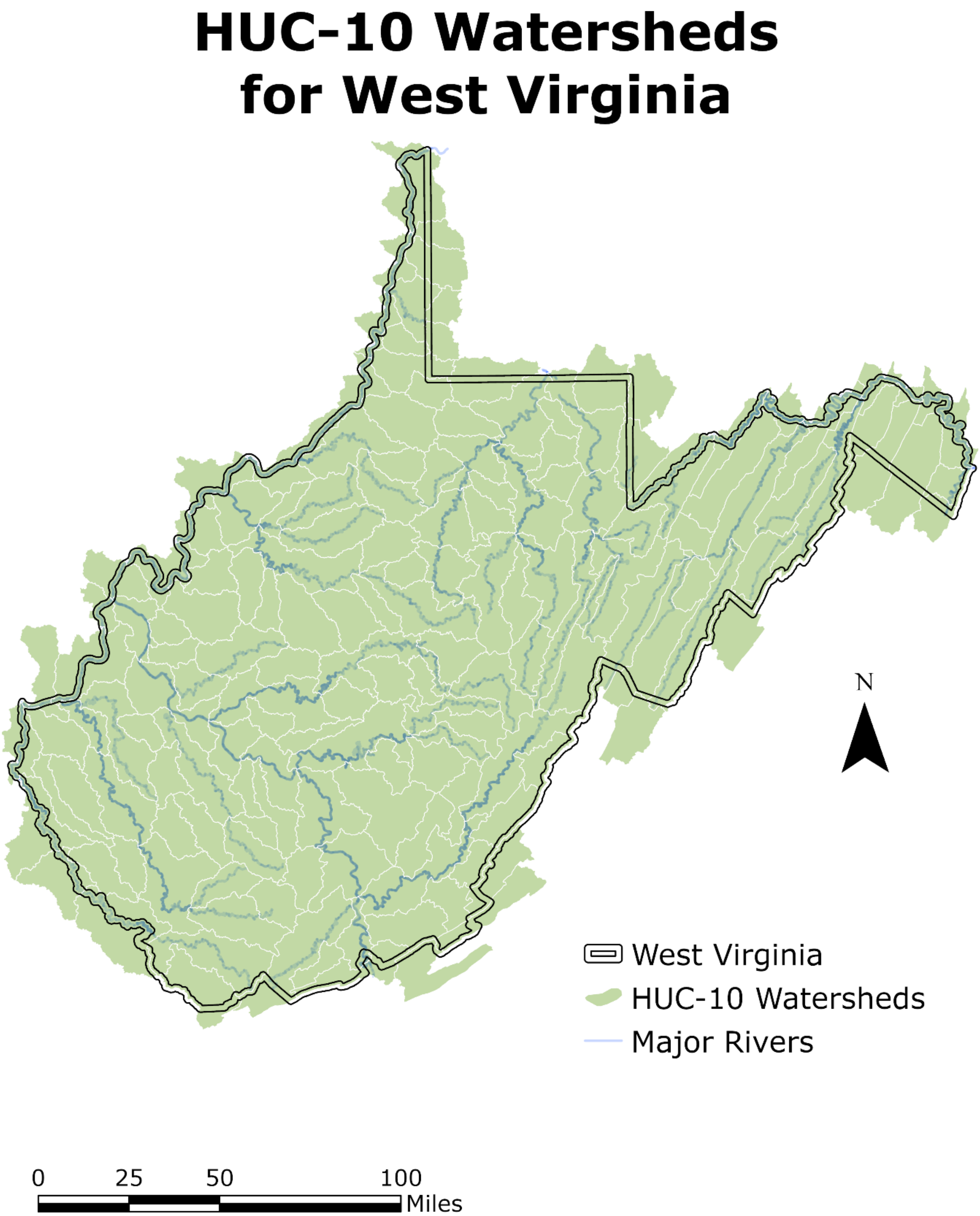
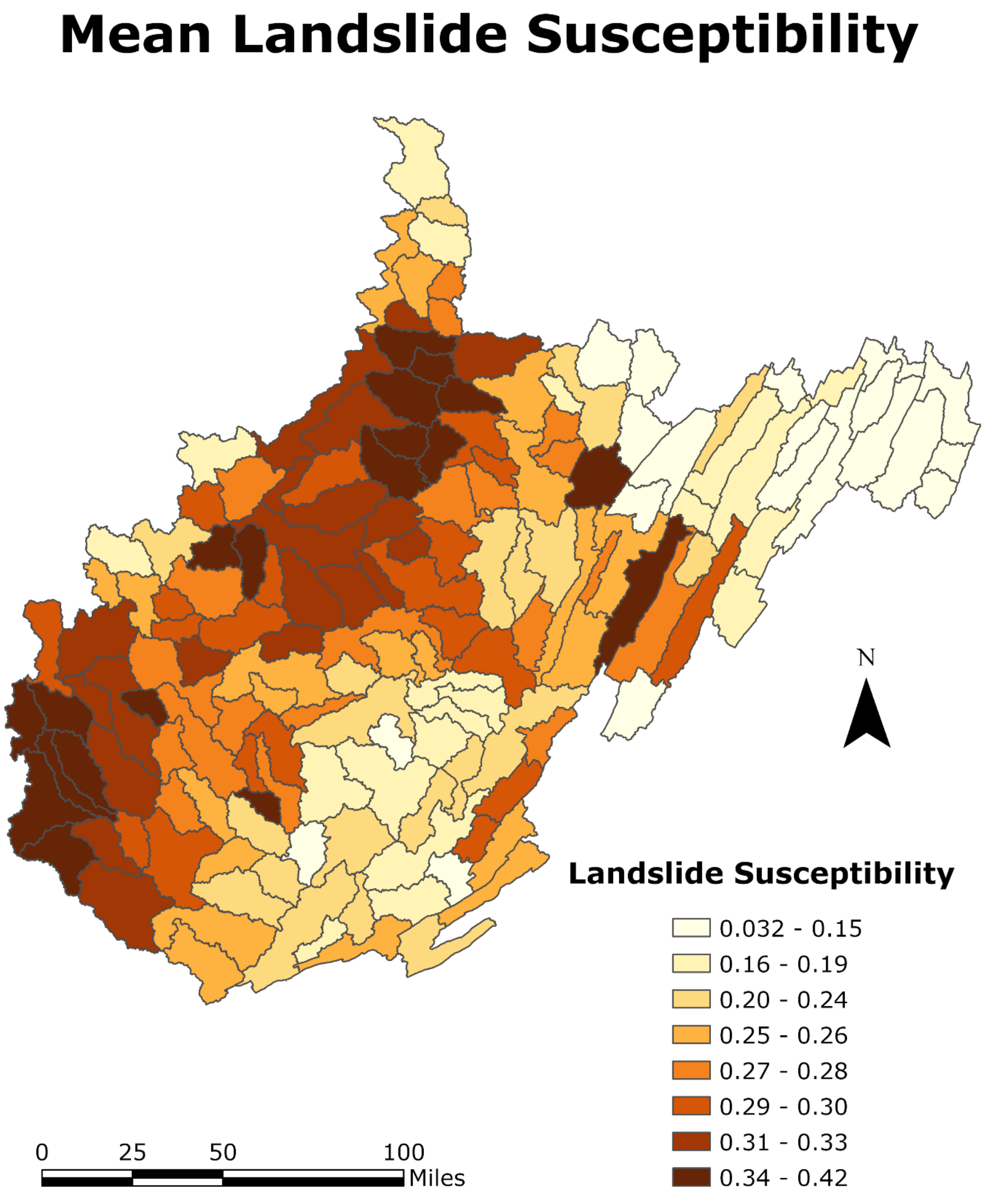
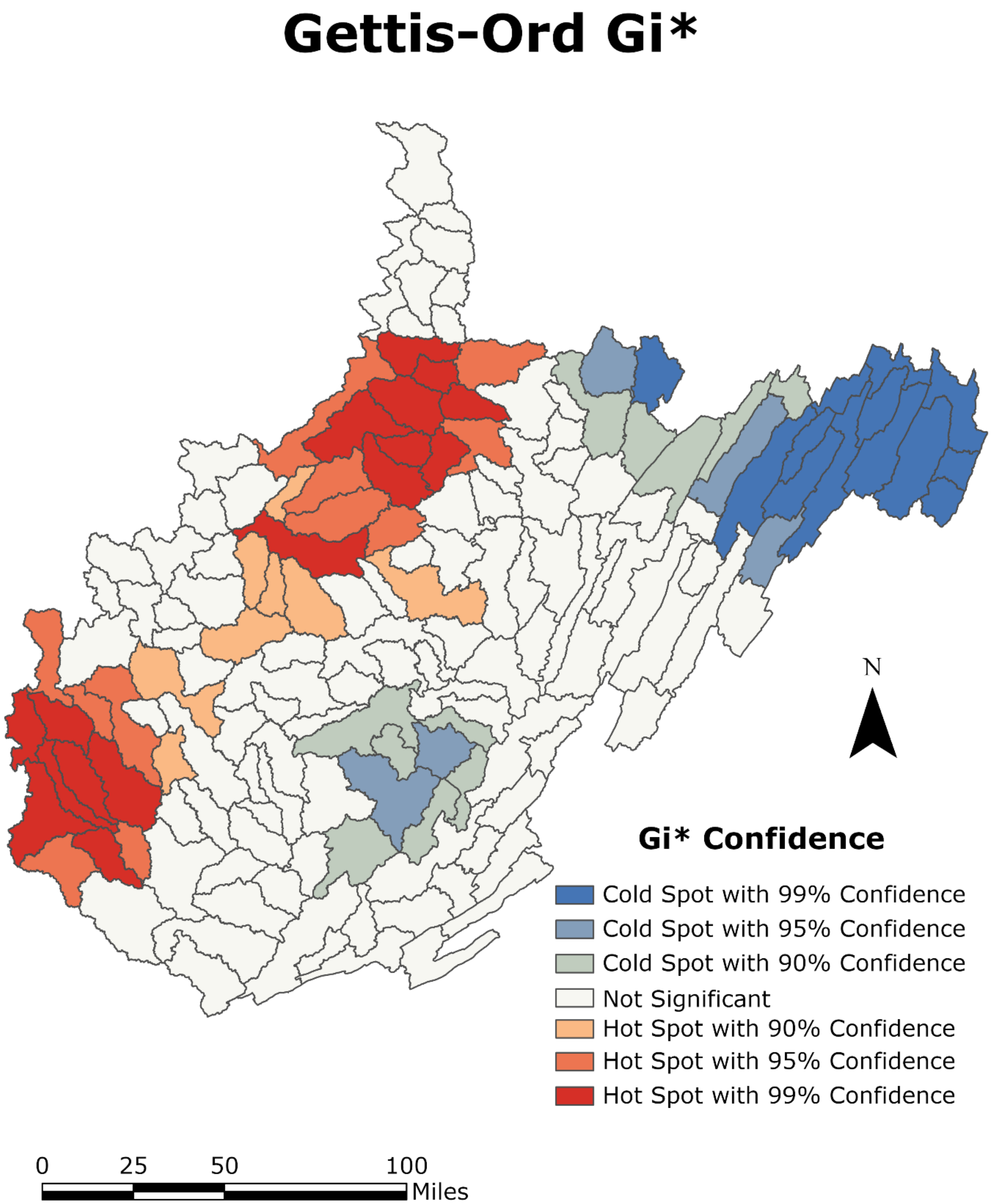
Maps and Figure



Map 1. HUC-10 Watersheds for West Virginia. Watershed divisions are shown outlined by West Virginia boundary and the major rivers within the state.



Map 2. Mean Landslide Susceptibility for West Virginia by HUC-10 Watershed. The mean of the landslide susceptibility by HUC-10 watershed divisions for the state of West Virginia. The map watersheds with a lower risk of landslides are indicated by yellow, while higher risks are indicated by red.



Map 3. Getis-Ord Gi* for West Virginia by HUC-10 Watershed. The map shows the results of the HUC-10 watershed level Getis-Ord Gi* with neighbors being defined as touching each other. The blues indicate clustering of low values, while reds indicate clustering of higher values.

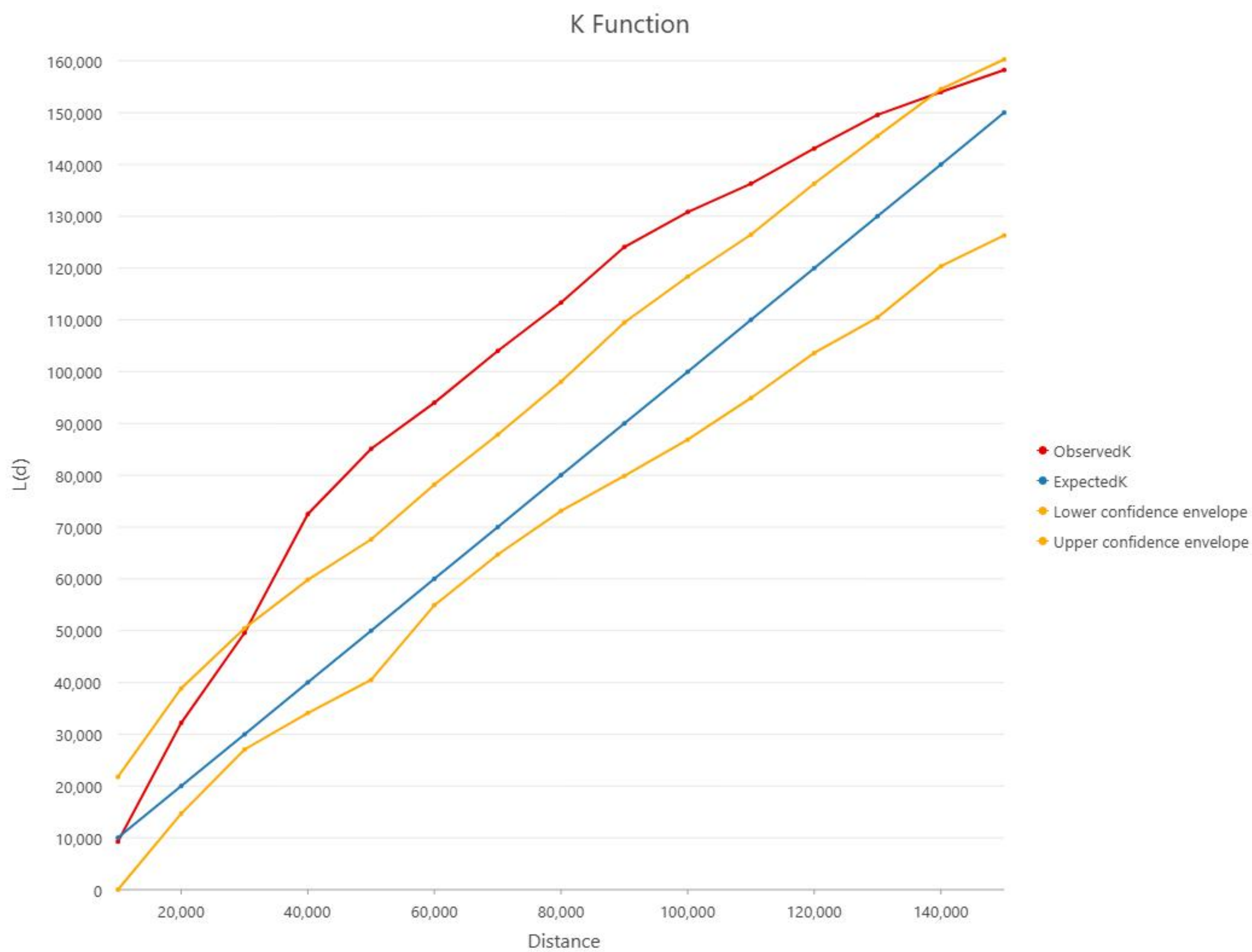


Figure 1. Ripley's K Function Graph. The graph shows the expected values of K, seen in blue, along with the parallel confidence envelopes around the expected values, seen in yellow. The actual observed values, seen in red, passed the upper confidence envelope at a distance of ~40 km and passed back through at a distance of ~140 km.